

# ABU HASNAT ABDULLAH

Senior AI Engineer · LLM · RAG · MLOps

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## PROFESSIONAL SUMMARY

Senior Software Engineer with **10+ years of professional experience**, including **6+ years** architecting production-grade AI systems for enterprise clients across the **Nordics, Sweden, and Bangladesh**. Ship LLM applications end-to-end — RAG pipelines, fine-tuned domain models (LoRA / QLoRA), and MLOps platforms on AWS and Azure — with measurable impact: lifted industrial CV detection accuracy **from 40% to 98%**, fine-tuned a local 3B LLM to **97% accuracy**, and led an **11-engineer team** across 6 concurrent enterprise projects. Obsessed with responsible-AI guardrails, evaluation discipline, and turning research-grade ideas into reliable production systems.

## TECHNICAL SKILLS

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| <b>Generative AI &amp; LLMs</b>     | OpenAI GPT, Anthropic Claude, Llama 3 / 3.2, Mistral, DeepSeek, Qwen; prompt engineering, RAG architectures, semantic chunking, evaluation frameworks, hallucination guardrails |
| <b>Fine-Tuning</b>                  | LoRA, QLoRA, PEFT, Unsloth, HuggingFace Transformers, domain adaptation                                                                                                         |
| <b>Vector &amp; Knowledge</b>       | PGVector, Chroma, Neo4j knowledge graphs, embedding optimization                                                                                                                |
| <b>ML &amp; Computer Vision</b>     | PyTorch, TensorFlow, scikit-learn, YOLOv11, OpenCV, OCR, Pandas, NumPy                                                                                                          |
| <b>MLOps &amp; Data Engineering</b> | Mage.ai (ETL / orchestration), MLflow, AzureML, model registry, feature stores, drift monitoring, CI/CD                                                                         |
| <b>Languages</b>                    | Python (primary), SQL, JavaScript, HTML, CSS                                                                                                                                    |
| <b>Backend &amp; APIs</b>           | FastAPI, Django, Flask, REST APIs, async services, microservices, Kafka, Redis                                                                                                  |
| <b>Databases</b>                    | PostgreSQL, MongoDB, InfluxDB (time-series)                                                                                                                                     |
| <b>Cloud &amp; DevOps</b>           | AWS (Lambda, EC2, ECS, S3, RDS, API Gateway), Azure, Docker                                                                                                                     |
| <b>Leadership &amp; Delivery</b>    | Cross-functional team leadership (11+ engineers), system design, technical mentorship, agile delivery, stakeholder management                                                   |

## PROFESSIONAL EXPERIENCE

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| <b>Senior Software Engineer</b> · Cefalo Bangladesh Ltd. · Sensa AS, Norway                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Nov 2024 — Present |
| <ul style="list-style-type: none"><li>Own AI engineering for Sensa, a Norwegian industrial-AI company — delivering LLM applications, computer-vision systems, time-series MLOps, and the underlying data platform end-to-end, with deep focus on evaluation, observability, and responsible-AI guardrails.</li><li><b>Unity AI</b> — Architected an LLM-powered conversational assistant that lets industrial operators query plant data in natural language. Designed a custom RAG pipeline combining a <b>Neo4j knowledge graph</b> with an <b>InfluxDB time-series store</b>, with grounded citations and data-governance rules that drive confident, source-backed answers and minimize hallucinations.</li></ul> |                    |
| › LLM, OpenAI, RAG, FastAPI, Neo4j, InfluxDB, PGVector, Pandas, SaaS, Azure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |

- **CV Analyser** — Shipped an AI résumé-parsing platform that extracts text from CVs and returns structured, ATS-ready output with role/skill categorization, plus a chatbot for recruiter Q&A over the parsed corpus. Fine-tuned **Llama 3.2, DeepSeek, and Qwen** with Unsloth (QLoRA), benchmarked trade-offs across model families, and reached **97% accuracy** on a local 3B model — matching cloud-API quality at a fraction of the cost.
  - › Llama 3.2, DeepSeek, Qwen, Unsloth, QLoRA / PEFT, HuggingFace, FastAPI, OCR, Python
- **Fish Food Pellet Detection** — Productionized a **YOLOv11-nano** computer-vision model that detects and counts fish-food pellets in real-time aquaculture footage, lifting detection accuracy **from 40% to 98%** and unlocking automated feed-rate optimization for sensor-equipped fish farms.
  - › YOLOv11, PyTorch, OpenCV, Python
- **Unity-MLOps** — Engineered an end-to-end anomaly-detection pipeline on time-series sensor data with automated training, model registry, deployment, and drift monitoring — turning manual log review into proactive, real-time alerts.
  - › FastAPI, Python, scikit-learn, AzureML, MLflow
- **Sensa Engine** — Built Sensa's internal **data-orchestration and feature platform** — the foundation powering Unity AI, Unity-MLOps, and Pellet Detection. Designed **Mage.ai ETL pipelines** to ingest sensor and external data into time-series and feature stores, with reusable transforms that serve both real-time anomaly detection and downstream RAG retrieval.
  - › Mage.ai, Python, FastAPI, InfluxDB, Neo4j, PGVector, Pandas, Azure

## Senior Software Engineer · Strativ AB · Sweden (Remote)

Apr 2021 — Jan 2024

- Led a cross-functional team of **11 engineers** across **6 concurrent enterprise projects** for top-tier Swedish clients — owning system design, backend architecture (Python, Django), code review, and delivery, while staying hands-on with API and AI feature development.
- **TourGPT** — Architected a SaaS RAG chatbot grounded in a custom tour knowledge base, delivering personalized recommendations and end-to-end order management. Owned prompt engineering, embedding strategy, and safety guardrails **from prototype to production**.
  - › LLM, OpenAI, RAG, Django, PostgreSQL, PGVector, SaaS, AWS
- **Investment Management System** — Designed and shipped a portfolio-analytics platform tracking performance, sales, and accounting across multiple portfolio companies, integrating third-party data sources via event-driven processing.
  - › Django, REST API, PostgreSQL, SaaS, Kafka, AWS
- **Health Check** — Delivered a digital primary-healthcare platform ingesting physical measurements and lab blood-test reports via third-party APIs, surfacing actionable health metrics for end users.
  - › Django, REST API, PostgreSQL, Kafka, NumPy, AWS

## Software Engineer · Circle Fintech Ltd.

Sep 2020 — Apr 2021

- Shipped Boarding Bay, an eKYC platform for **Jamuna Bank**, modernizing customer onboarding and accelerating business financial transaction processing across branches.
- **Boarding Bay** — Built 3 production-grade microservices (NID OCR, face detection, billing) that automated identity verification and significantly reduced manual onboarding time.
  - › FastAPI, Flask, Django, REST API, Microservices, TensorFlow, PyTorch, Pandas, NumPy

## Junior Research Analyst · Economic Research Group

Apr 2020 — Aug 2020

- Analyzed mobile-app data for Romoni (on-demand beauty marketplace) to drive sales growth and forecast demand using regression models, informing pricing and marketing decisions.

› Python, PyTorch, NumPy, Pandas, Matplotlib, Regression

**Founder & CEO** · Shikhte Chai (Educational Initiative)

Apr 2016 — Jan 2020

- Founded and scaled an educational initiative in Bangladesh, leading strategy, operations, and program design to upskill students in technology and entrepreneurship.

## EDUCATION

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**B.Sc. in Computer Science & Engineering** · North South University, Bangladesh

2015 — 2019

- CGPA: 3.07 / 4.00
- Relevant Coursework: Artificial Intelligence, Pattern Recognition & Neural Networks, Data Structures, Design and Analysis of Algorithms, Probability and Statistics, Microeconomics.

## CERTIFICATIONS

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**Neo4j GraphAcademy**

2025

- **Neo4j Fundamentals** · **Cypher Fundamentals** · **Neo4j & Generative AI Fundamentals** — graph databases, Cypher query language, and grounded RAG with knowledge graphs.

**Data Science** · **Prof. Jennifer Widom, Stanford University**

2024

- Machine learning (regression, classification, clustering), data manipulation, mining, and visualization with Tableau, Python, and R.

**Business Analysis & Prototyping** · **LICT (Bangladesh Govt.) & BracIT**

2019

- 3-month program in requirements analysis, strategic planning, design thinking, process improvement, and stakeholder communication.

## RESEARCH PUBLICATIONS

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- **Predicting the Result of a Cricket Match by Applying Data Mining Techniques** (2020) — Applied Recursive Feature Elimination, ZeroR, Decision Tree, Random Forest, and XGBoost to predict match outcomes. Published on ResearchGate.